

(AI) to compare the patient’s DNA sequence against a standard human or a population-specific reference. This comparison flags minute errors or variants—like a misspelled word in a user’s manual—that cause disease or predict how a patient will respond to a drug.

THE DEMAND SURGE

At tertiary cancer centres in Bengaluru, Hyderabad, Mumbai, Pune and Gurugram, oncologists notice that patients and clinicians are ordering multi-gene panels at volumes unimaginable even five years ago. “Oncology continues to drive the highest demand for genomic testing in India,” says Dr Shrinidhi Nathany, consultant of Molecular Haematology and Oncology at Fortis Memorial Research Centre in Gurugram, who works at the intersection of targeted therapies and next-generation sequencing.

62 | She sees genomic profiling becoming so routine that therapy selection without molecular input would be incomplete.

“We use it to guide therapy, detect resistance and monitor minimal residual disease,” she says, noting a marked increase in adoption every quarter.

Rare-disease diagnostics, once scattered across a handful of paediatric geneticists, have also expanded. With clinical exome sequencing—a DNA test for rare genetic disorders—becoming more accessible and reporting improving

quality, paediatricians are increasingly turning to genomics when investigations for delays or undiagnosed syndromes reach a dead end.

“Rare disease diagnostics are rising steadily,” Dr. Nathany says, adding that improved turnaround times make it easier for clinicians to integrate genomic findings into management decisions.

Reproductive genomics forms a third, rapidly growing pillar. Fertility clinics, maternal-foetal medicine specialists and high-risk obstetrics units increasingly rely on Non-Invasive Prenatal Testing (NIPT), carrier screening and pre-implantation genetic testing.

Genome-based pregnancy management, once limited to tertiary centres, is now extending into private chains that treat genomics as essential for modern prenatal care.

“The demand for genomic testing in India has seen a major surge over the last five years,” says Anuradha Acharya, CEO of Mapmygenome and Microbiome Insights. Part of this, she explains, is rooted in heightened post-pandemic health awareness. But the more structural change is hospitals integrating genomics into preventive packages, allowing consumers to access carrier status, cardiometabolic risk and pharmacogenetic information.

REDUCING UNCERTAINTY

Acharya notes that clinical confidence is rising as AI-validated

THE GENOME STORY

- 1 Genomic testing demand is rising with increasing cancer and rare-disease burden
- 2 Hospitals and diagnostic chains are rapidly expanding genomic labs and comprehensive genomic profiling panels
- 3 India-specific datasets from Bharat Cancer Genome Atlas and tribal genome projects are strengthening precision medicine
- 4 Global investors such as Novo, Maj Invest and LeapFrog are stepping up funding in Indian genomics
- 5 India’s estimated ₹4,876-crore genomics market is growing at a CAGR of 18%

MARKET SIZE (2024)

\$550
MILLION

18%
Projected
CAGR (2024–
2030)

1.5
MILLION CASES
Annual Cancer
incidence

70
MILLION
Rare genetic
disorder burden

KEY DRIVERS
Sequencing
cost decline, oncology
demand, datasets